Dual-Site Enterprise Network Design

By: Andres Jose L. Tadeo

A Little About This Project

Hi, I am Andres. I built this project to challenge myself and prepare for designing and configuring networks similar to those used in real-world enterprise environments.

My goal was to see whether I could build a secure, professional-grade network from scratch that connects two separate sites while implementing routing, VPNs, security policies, and essential network services.

Packet Tracer Version Used: v9.0.0.0810

Download: https://www.netacad.com/articles/news/download-cisco-packet-tracer

Project Objectives

Objectives

- Hierarchical Enterprise Design
- Secure Site-to-Site Connectivity
- Dynamic Routing
- Public and Private Services
- Redundancy and High Availability
- Layered Security Architecture

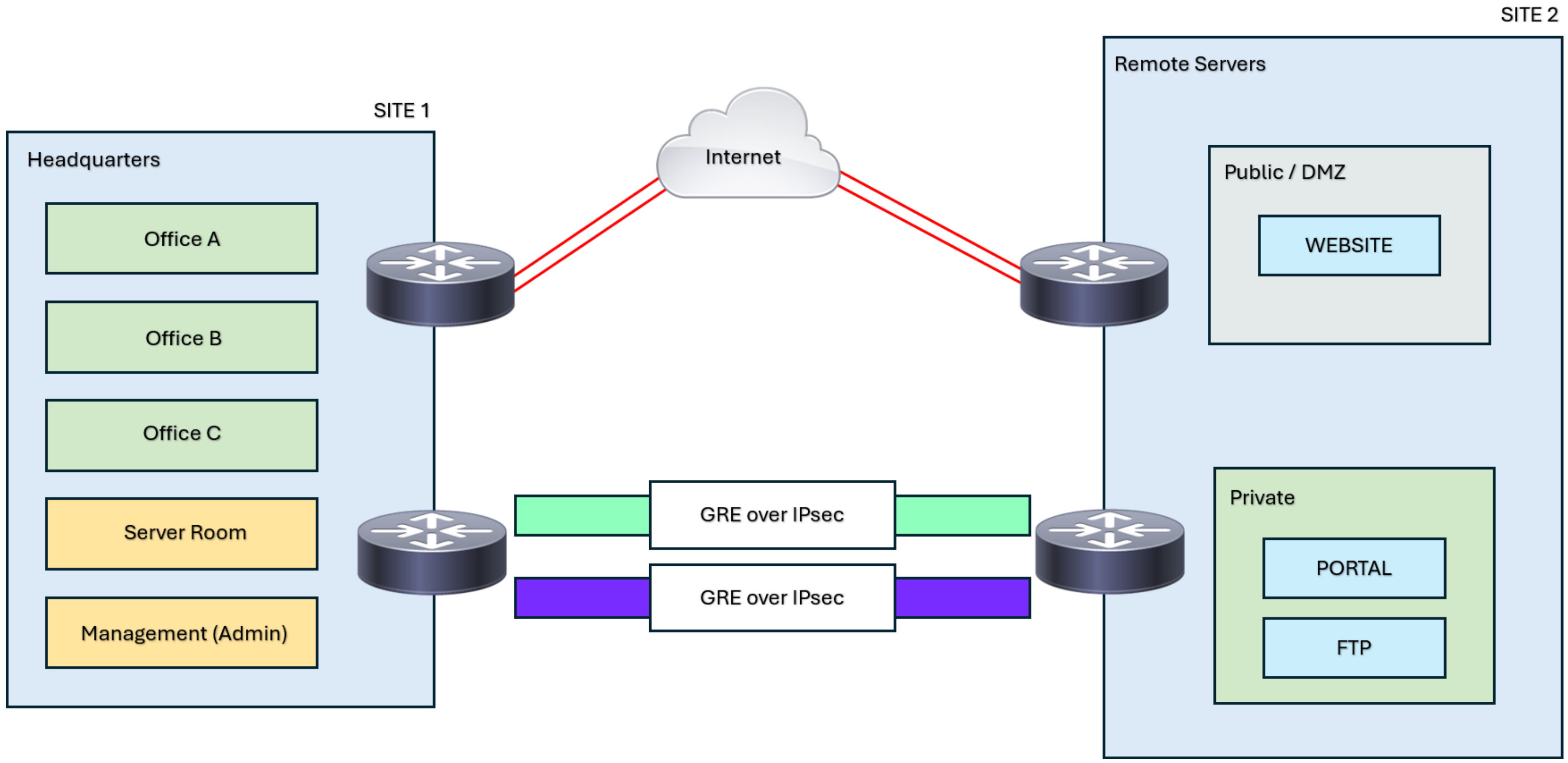
Topology Overview

The network topology represents a dual-site enterprise WAN architecture connecting the Headquarters and Remote Site through redundant GRE over IPsec tunnels across WAN connections.

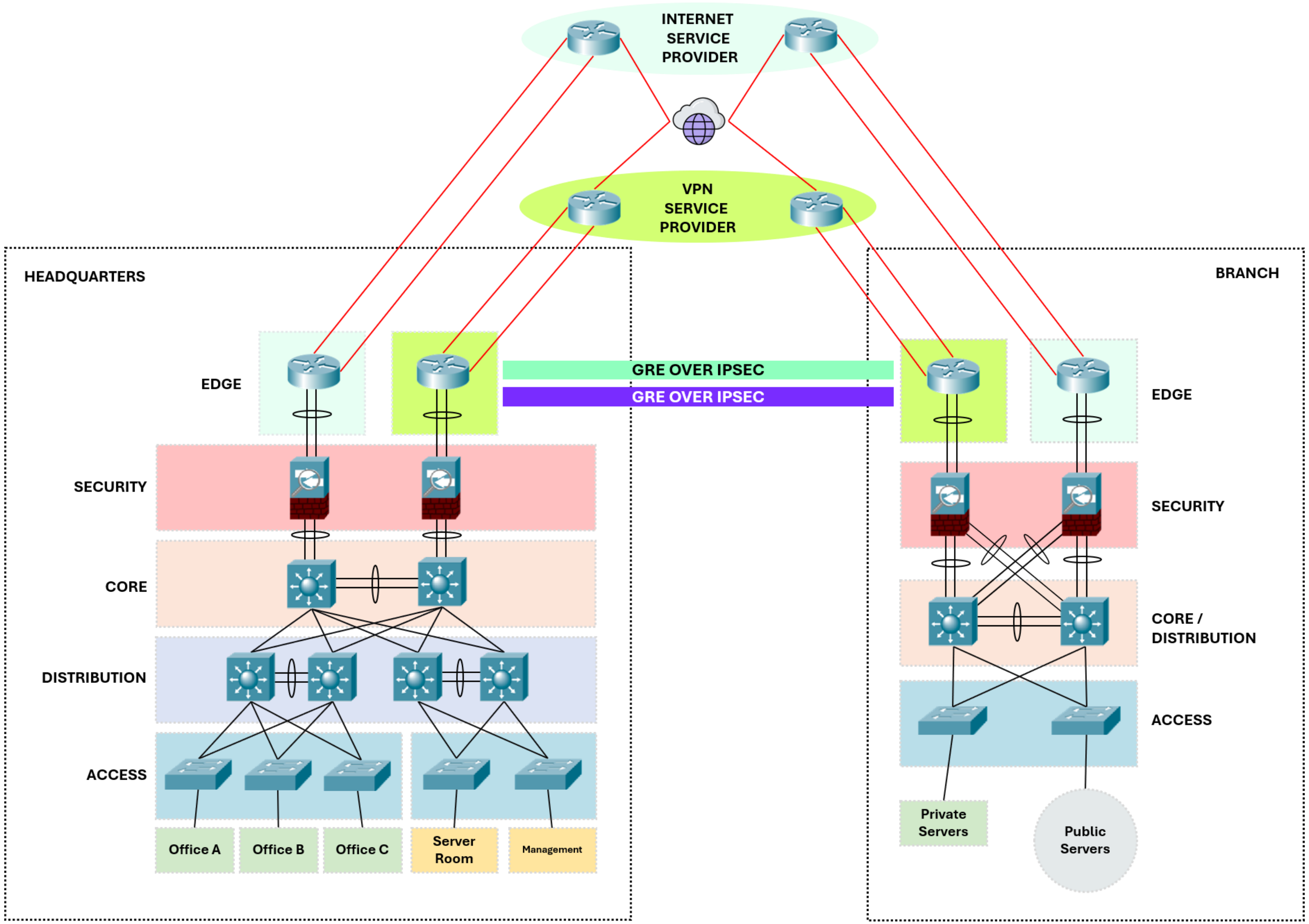
The design incorporates a three-tier campus architecture, VLAN segmentation, Layer 3 switching, dynamic routing, gateway redundancy, and perimeter firewalls to provide scalability, resiliency, and secure inter-site communication.

Dedicated edge routers separate Internet access and VPN functions, with one router handling Internet connectivity and the other managing GRE over IPsec tunnel operations.

Simplified Topology View



Full Topology View



Headquarters

The Headquarters network uses a ***Three-Tier Architecture*** (Access, Distribution, and Core) to provide scalability, redundancy, and efficient traffic management.

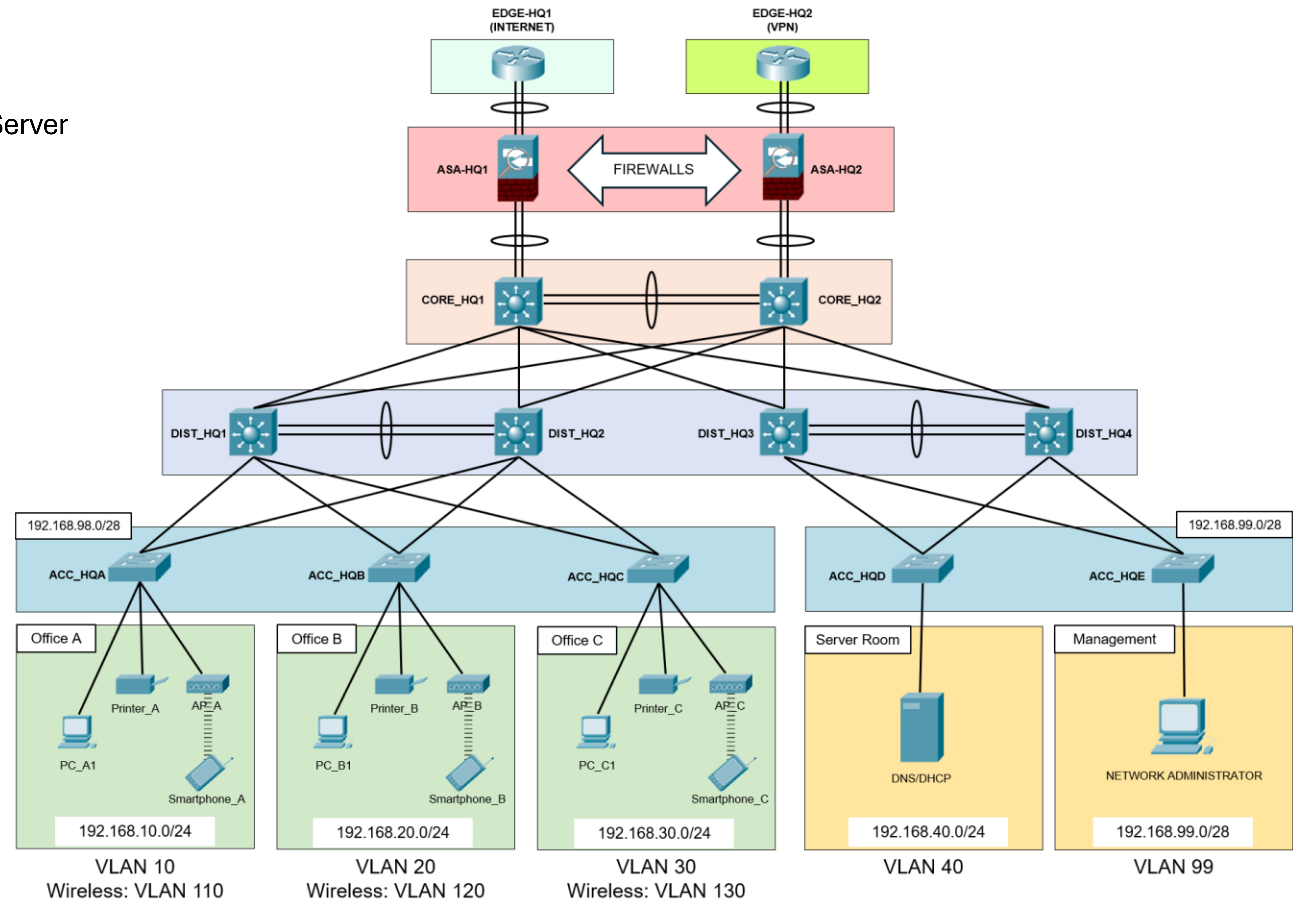
The network is segmented into Office A, Office B, Office C, Server Room, and Management VLANs, with a centralized server providing DHCP and DNS services.

Components

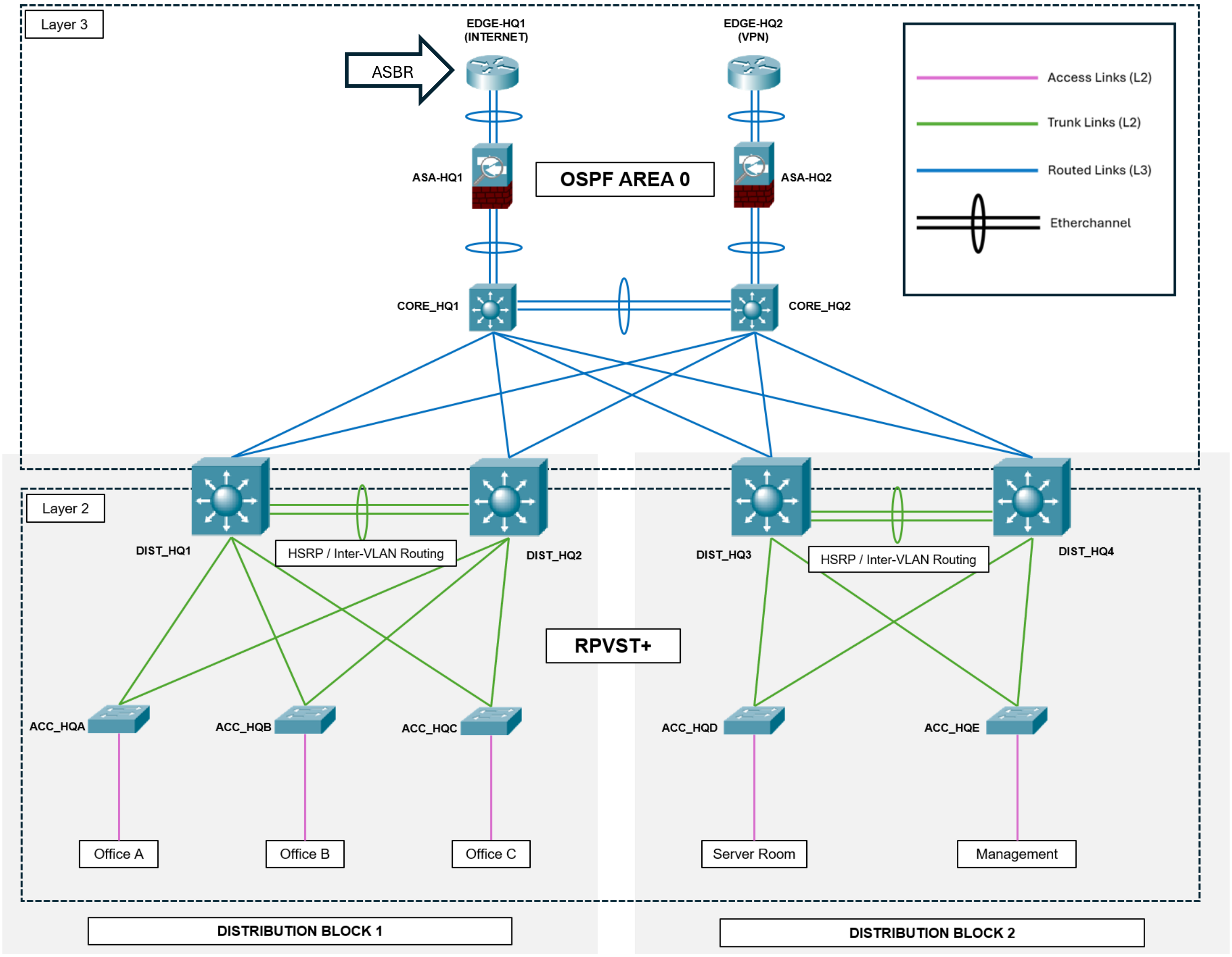
- Access Switches
- Distribution Switches
- Core Switches
- Dual Firewalls
- Dual Edge Routers

Design Features

- Layered architecture
- EtherChannels
- OSPF routing
- Redundant paths
- HSRP
- RPVST+



Headquarters Network Architecture



Branch (Remote Servers)

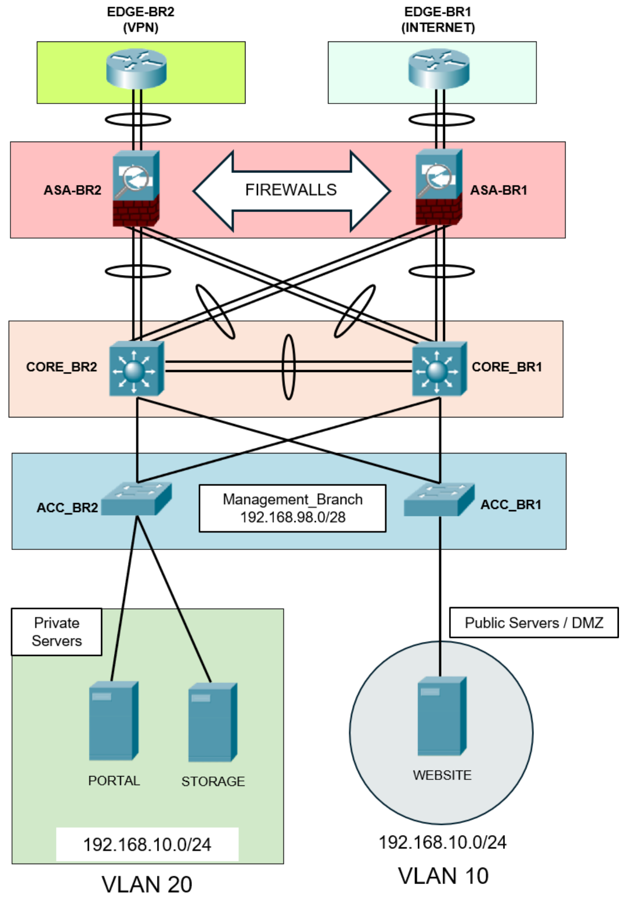
The **Remote Servers** network is designed to securely host services by separating public-facing and internal resources. It uses a **Two-tier Collapsed Core Architecture**, with core and distribution functions combined on the CORE_BR1 and CORE_BR2 switches.

Components

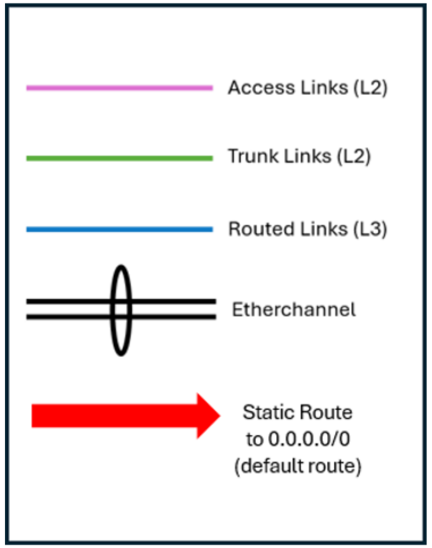
- Core Switches
- Access Switches
- Internal Server Segment
- DMZ Network
- Dual Firewalls
- Dual Edge Routers

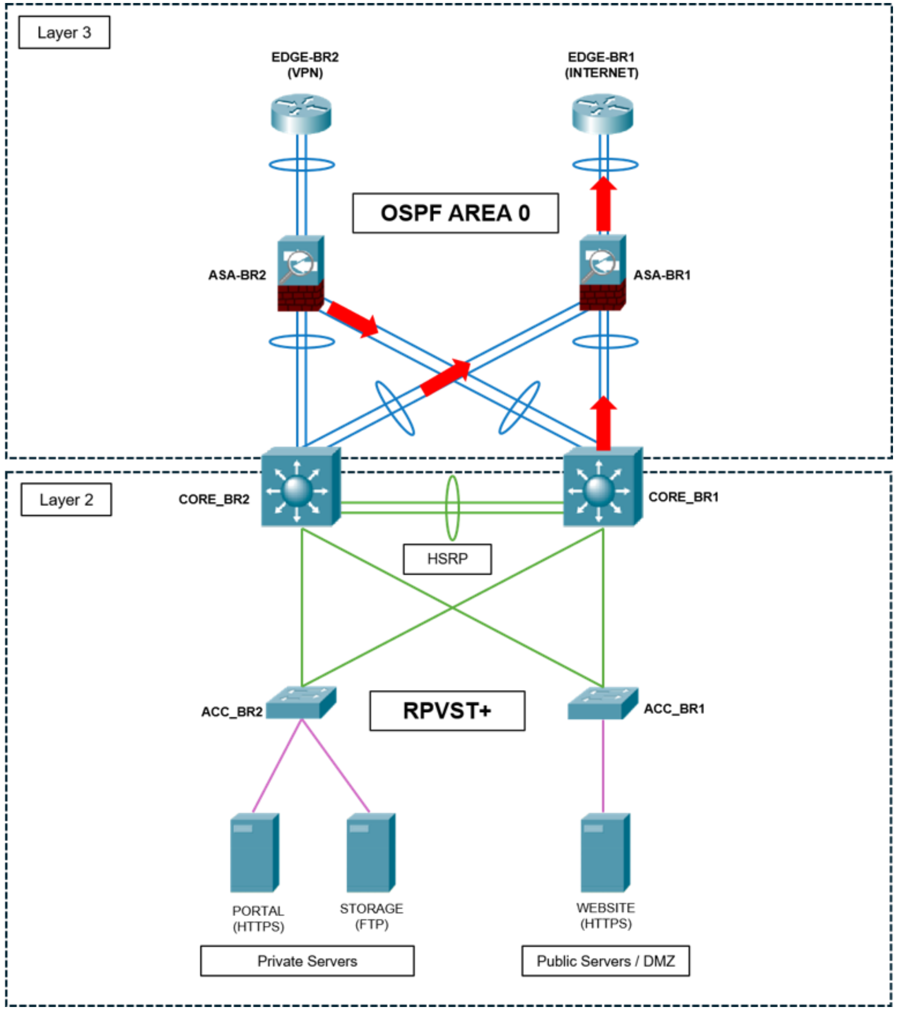
Design Features

- Secure server hosting
- DMZ isolation
- Layered architecture
- EtherChannels
- OSPF routing
- RPVST+



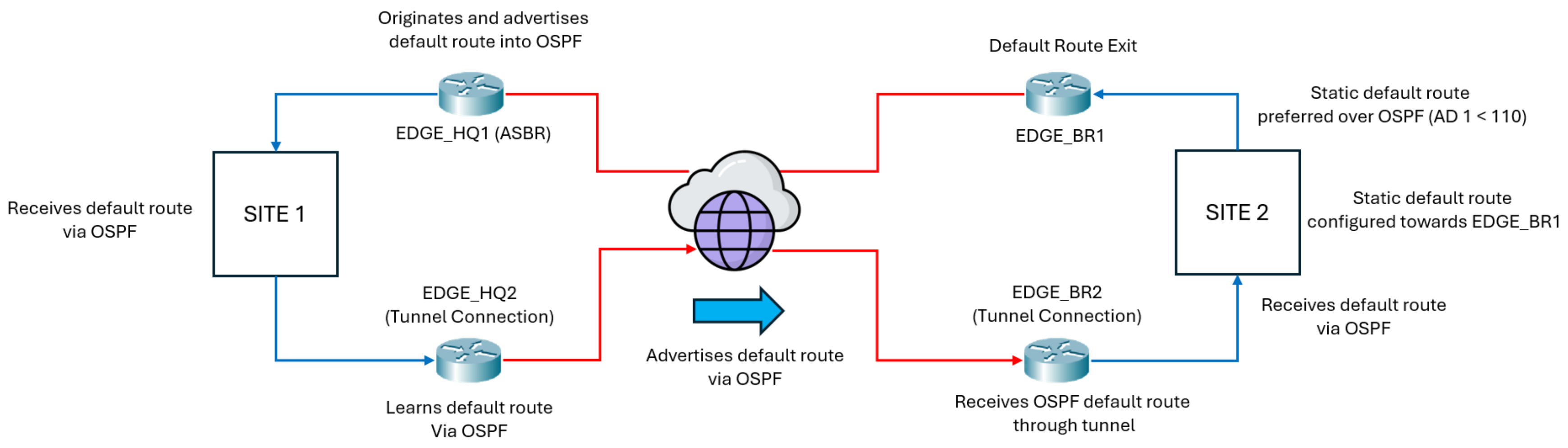
Branch Network Architecture





Why use Static Default Route?

- Packet Tracer supports only a limited set of Cisco IOS features and does not fully support advanced OSPF default-route behavior
- The `default-information originate <metric-type> command` is not available in Packet Tracer.
- Policy-Based Routing (PBR) is also not supported.
- A static default route has an Administrative Distance of 1, making it more preferred than an OSPF route with an Administrative Distance of 110.
- The result is predictable route selection while preserving OSPF route propagation across both sites.



VLAN Architecture

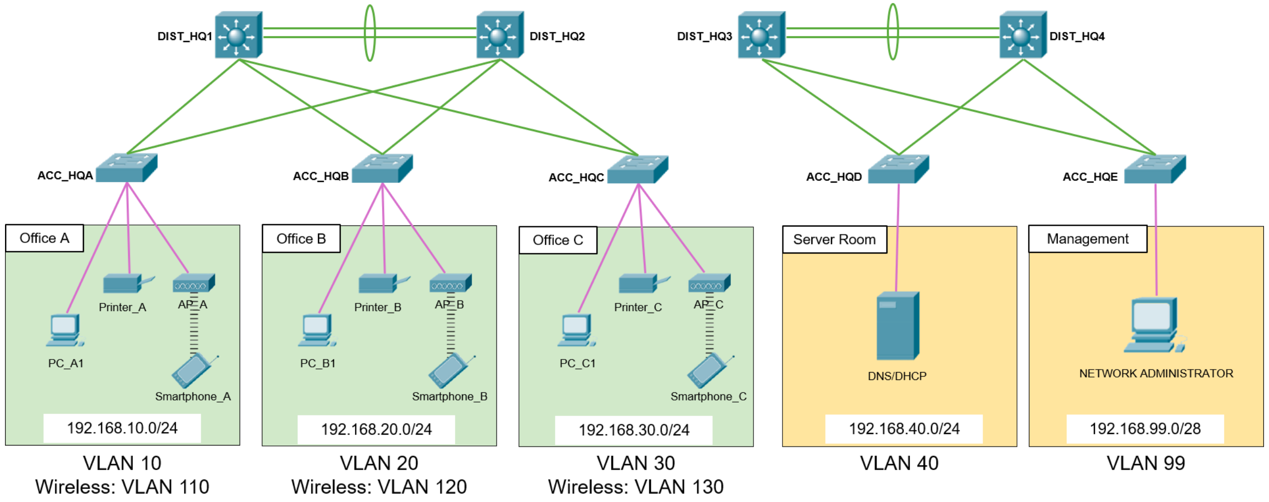
HEADQUARTERS					
VLAN	**Name**	**Subnet**	**Gateway (VIP)**	**HSRP Active**	**HSRP Standby**
10	Office A	192.168.10.0/24	192.168.10.1	DIST_HQ1	DIST_HQ2
20	Office B	192.168.20.0/24	192.168.20.1	DIST_HQ1	DIST_HQ2
30	Office C	192.168.30.0/24	192.168.30.1	DIST_HQ1	DIST_HQ2
40	Server_Room	192.168.40.0/24	192.168.40.1	DIST_HQ3	DIST_HQ4
99	Management_12	192.168.98.0/28	192.168.98.1	DIST_HQ2	DIST_HQ1
	Management_34	192.168.99.0/28	192.168.99.1	DIST_HQ4	DIST_HQ3
110	AP_A	192.168.110.0/24	192.168.110.1	DIST_HQ2	DIST_HQ1
120	AP_B	192.168.120.0/24	192.168.120.1	DIST_HQ2	DIST_HQ1
130	AP_C	192.168.130.0/24	192.168.130.1	DIST_HQ2	DIST_HQ1
1001	Native VLAN	X	X	X	X
BRANCH					
VLAN	**Name**	**Subnet**	**Gateway (VIP)**	**HSRP Active**	**HSRP Standby**
10	Public_Servers	172.16.10.0/28	172.16.10.1	DIST_HQ1	DIST_HQ2
20	Private_Servers	172.16.20.0/28	172.16.20.1	DIST_HQ2	DIST_HQ1
99	Management_Branch	172.16.99.0/28	172.16.99.1	DIST_HQ2	DIST_HQ1
1001	Native VLAN	X	X	X	X

Layer 2 Architecture

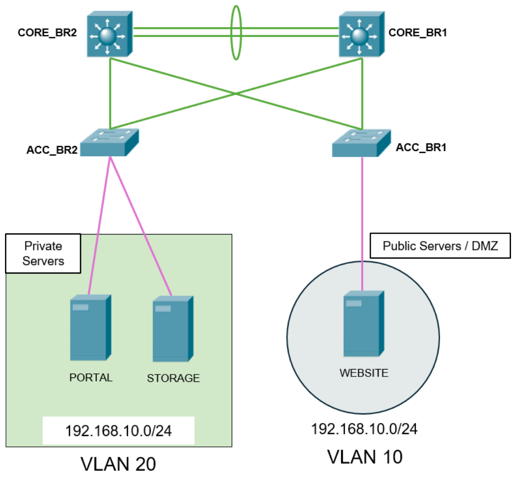
Technologies
- EtherChannel
- Rapid PVST+
- PortFast
- BPDU Guard

Benefits
- Redundancy
- Fast convergence
- Loop prevention

HEADQUARTERS



Branch



Spanning Tree Protocol Design (HQ)

Rapid Per-VLAN Spanning Tree Plus (RPVST+) is deployed across the switching infrastructure to provide fast Layer 2 convergence, prevent switching loops, and support redundant network paths.

Root Bridge election is aligned with HSRP active gateway assignments to ensure optimal traffic flow

Additional STP Features:

➢ **PortFast**

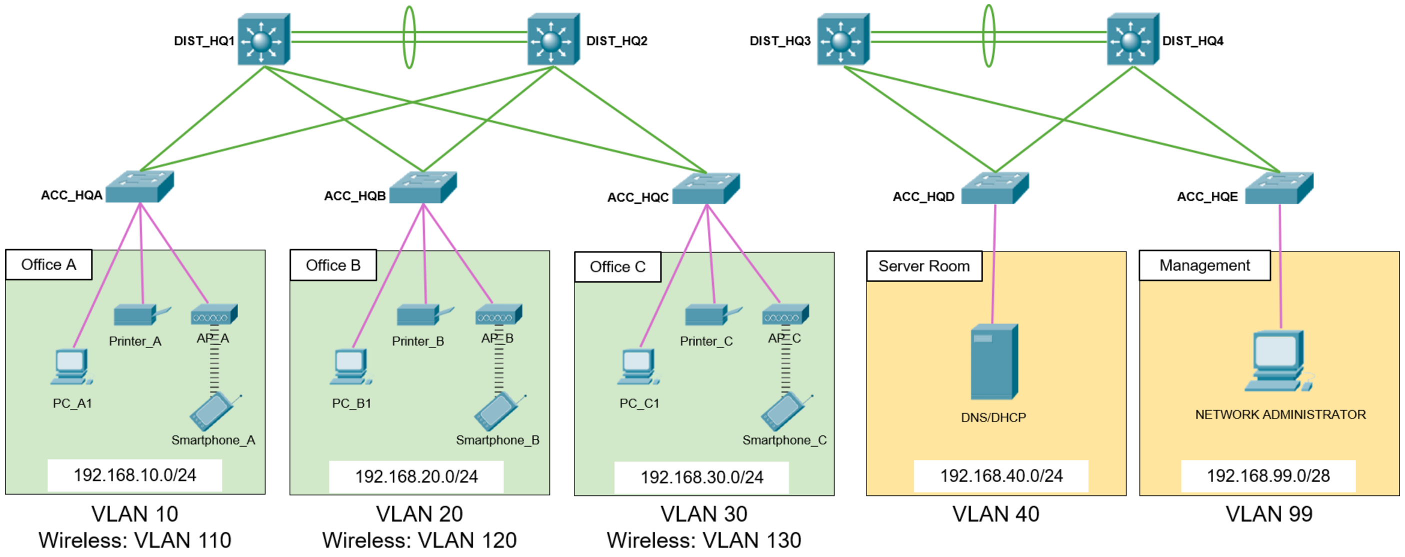
Configured on all access ports connected to:

- PCs
- Printers
- Servers
- End-user devices

➢ **BPDU Guard**

Enabled on all PortFast interfaces.

ROOT BRIDGE PLACEMENT		
VLAN	**Primary Root**	**Secondary Root**
10, 20, 30	DIST_HQ1	DIST_HQ2
99, 110, 120, 130	DIST_HQ2	DIST_HQ1
40	DIST_HQ3	DIST_HQ4
99	DIST_HQ4	DIST_HQ3



Spanning Tree Protocol Design (Branch)

Rapid Per-VLAN Spanning Tree Plus (RPVST+) is deployed across the switching infrastructure to provide fast Layer 2 convergence, prevent switching loops, and support redundant network paths.

Root Bridge election is aligned with HSRP active gateway assignments to ensure optimal traffic flow

Additional STP Features:

➢ **PortFast**

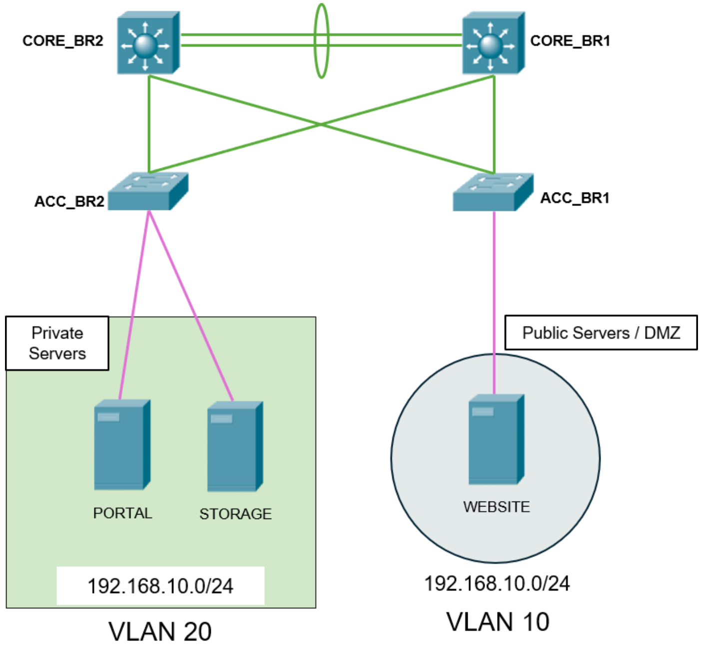
Configured on all access ports connected to:

- Private Servers
- Public Servers

➢ **BPDU Guard**

Enabled on all PortFast interfaces.

ROOT BRIDGE PLACEMENT		
VLAN	**Primary Root**	**Secondary Root**
10	CORE_BR1	CORE_BR2
20, 99	CORE_BR2	CORE_BR1

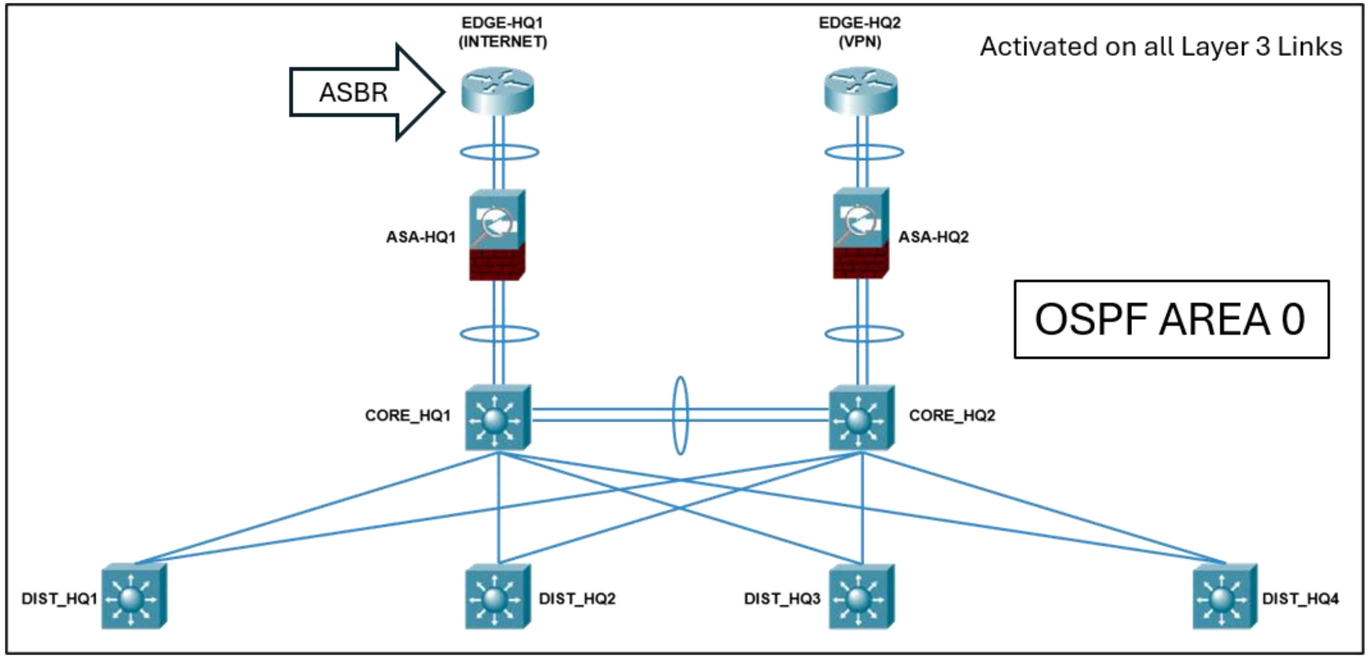


OSPF Design (HQ)

Open Shortest Path First (Single Area) is selected as the dynamic routing protocol to provide efficient connectivity between the Layer 3 Switches, Firewalls and Edge Routers, it automatically calculates the best paths for traffic without manual route configuration.

To maintain stability, VLAN and Loopback interfaces are configured as `passive-interface` to prevent unnecessary adjacencies, while primary and backup default routes (0.0.0.0/0) are injected at EDGE_HQ1 (**Autonomous System Boundary Router**) to manage all outbound traffic.

Device	Router-ID
DIST_HQ1	10.255.255.1
DIST_HQ2	10.255.255.2
DIST_HQ3	10.255.255.3
DIST_HQ4	10.255.255.4
CORE_HQ1	10.255.255.10
CORE_HQ2	10.255.255.11
ASA-HQ1	10.255.255.15
ASA-HQ2	10.255.255.16
EDGE_HQ1	10.255.255.21
EDGE_HQ2	10.255.255.22



Routing Verification (HQ)

This is a sample routing table from **CORE_HQ1** that shows **OSPF** has successfully exchanged and installed routes across the network, providing reachability to all required destinations.

Multiple equal-cost paths are available for several networks, enabling **ECMP** (Equal-Cost Multi-Path) to improve overall network performance and availability.

ECMP provides:

- Redundancy
- Load Balancing
- Scalability
- Failover
- Efficiency
- Reliability

```
CORE_HQ1#show ip route ospf
     10.0.0.0/8 is variably subnetted, 27 subnets, 2 masks
O       10.10.20.0 [110/2] via 10.10.10.1, 00:00:37, GigabitEthernet1/0/1
                   [110/2] via 10.5.5.2, 00:00:37, Port-channel2
O       10.10.40.0 [110/2] via 10.10.30.1, 00:00:37, GigabitEthernet1/0/2
                   [110/2] via 10.5.5.2, 00:00:37, Port-channel2
O       10.10.60.0 [110/2] via 10.10.50.1, 00:00:37, GigabitEthernet1/0/3
                   [110/2] via 10.5.5.2, 00:00:37, Port-channel2
O       10.10.80.0 [110/2] via 10.10.70.1, 00:00:37, GigabitEthernet1/0/4
                   [110/2] via 10.5.5.2, 00:00:37, Port-channel2
O       10.20.10.4 [110/2] via 10.20.10.2, 00:00:37, Port-channel1
O       10.20.20.0 [110/2] via 10.5.5.2, 00:00:37, Port-channel2
O       10.20.20.4 [110/3] via 10.5.5.2, 00:00:37, Port-channel2
O       10.255.255.1 [110/2] via 10.10.10.1, 00:01:07, GigabitEthernet1/0/1
O       10.255.255.2 [110/2] via 10.10.30.1, 00:01:07, GigabitEthernet1/0/2
O       10.255.255.3 [110/2] via 10.10.50.1, 00:01:07, GigabitEthernet1/0/3
O       10.255.255.4 [110/2] via 10.10.70.1, 00:01:07, GigabitEthernet1/0/4
O       10.255.255.12 [110/2] via 10.5.5.2, 00:00:37, Port-channel2
O       10.255.255.21 [110/3] via 10.20.10.2, 00:00:37, Port-channel1
O       10.255.255.22 [110/4] via 10.5.5.2, 00:00:37, Port-channel2
O    192.168.10.0 [110/2] via 10.10.10.1, 00:01:07, GigabitEthernet1/0/1
                  [110/2] via 10.10.30.1, 00:01:07, GigabitEthernet1/0/2
O    192.168.20.0 [110/2] via 10.10.10.1, 00:01:07, GigabitEthernet1/0/1
                  [110/2] via 10.10.30.1, 00:01:07, GigabitEthernet1/0/2
O    192.168.30.0 [110/2] via 10.10.10.1, 00:01:07, GigabitEthernet1/0/1
                  [110/2] via 10.10.30.1, 00:01:07, GigabitEthernet1/0/2
O    192.168.40.0 [110/2] via 10.10.50.1, 00:01:07, GigabitEthernet1/0/3
                  [110/2] via 10.10.70.1, 00:01:07, GigabitEthernet1/0/4
     192.168.98.0/28 is subnetted, 1 subnets
O       192.168.98.0 [110/2] via 10.10.10.1, 00:01:07, GigabitEthernet1/0/1
                     [110/2] via 10.10.30.1, 00:01:07, GigabitEthernet1/0/2
     192.168.99.0/28 is subnetted, 1 subnets
O       192.168.99.0 [110/2] via 10.10.50.1, 00:01:07, GigabitEthernet1/0/3
O    192.168.110.0 [110/2] via 10.10.10.1, 00:01:07, GigabitEthernet1/0/1
                   [110/2] via 10.10.30.1, 00:01:07, GigabitEthernet1/0/2
O    192.168.120.0 [110/2] via 10.10.10.1, 00:01:07, GigabitEthernet1/0/1
                   [110/2] via 10.10.30.1, 00:01:07, GigabitEthernet1/0/2
O    192.168.130.0 [110/2] via 10.10.10.1, 00:01:07, GigabitEthernet1/0/1
                   [110/2] via 10.10.30.1, 00:01:07, GigabitEthernet1/0/2
O*E2 0.0.0.0/0 [110/1] via 10.20.10.2, 00:00:37, Port-channel1
```

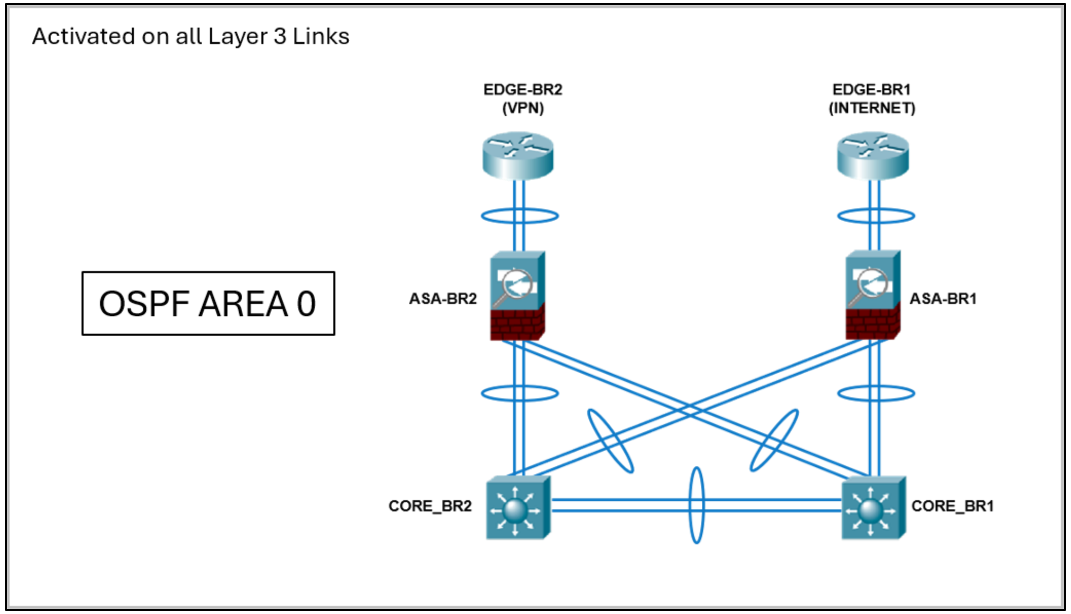
Default Route learned from ASBR

OSPF Design (Branch)

Open Shortest Path First (Single Area) is selected as the dynamic routing protocol to provide efficient connectivity between the Layer 3 Switches, Firewalls and Edge Routers, it automatically calculates the best paths for traffic without manual route configuration.

To maintain stability, VLAN and Loopback interfaces are configured as `passive-interface` to prevent unnecessary adjacencies

Device	Router-ID
CORE_BR1	10.255.255.31
CORE_BR1	10.255.255.32
ASA-BR1	10.255.255.35
ASA-BR2	10.255.255.36
EDGE_BR1	10.255.255.41
EDGE_BR2	10.255.255.42



Routing Verification (Branch)

This is a sample routing table from CORE_BR1 that shows OSPF has successfully exchanged and installed routes across the network, providing reachability to all required destinations.

A static route is configured towards the external router was configured.

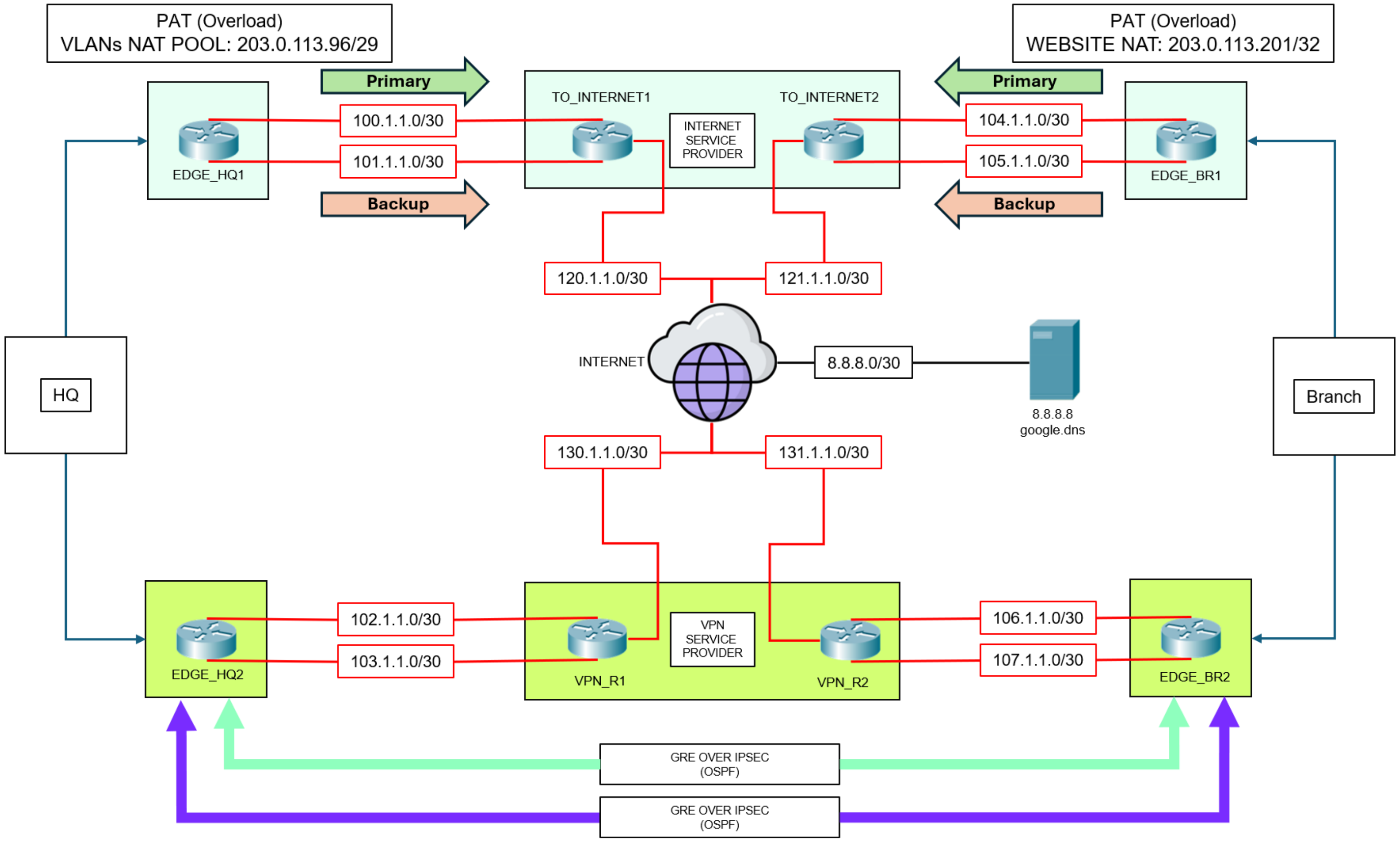
```
CORE_BR1#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.30.10.2 to network 0.0.0.0

     10.0.0.0/8 is variably subnetted, 12 subnets, 2 masks
C       10.30.10.0/30 is directly connected, Port-channel1
L       10.30.10.1/32 is directly connected, Port-channel1
O       10.30.10.4/30 [110/2] via 10.30.10.2, 00:01:12, Port-channel1
O       10.30.10.8/30 [110/51] via 172.16.99.3, 00:01:12, Vlan99
                      [110/51] via 10.30.10.2, 00:01:12, Port-channel1
O       10.30.20.0/30 [110/2] via 172.16.99.3, 00:01:12, Vlan99
O       10.30.20.4/30 [110/3] via 172.16.99.3, 00:01:02, Vlan99
C       10.30.20.8/30 is directly connected, Port-channel3
L       10.30.20.9/32 is directly connected, Port-channel3
C       10.255.255.31/32 is directly connected, Loopback0
O       10.255.255.32/32 [110/2] via 172.16.99.3, 00:01:12, Vlan99
O       10.255.255.41/32 [110/3] via 10.30.10.2, 00:01:02, Port-channel1
O       10.255.255.42/32 [110/4] via 172.16.99.3, 00:01:02, Vlan99
     172.16.0.0/16 is variably subnetted, 6 subnets, 2 masks
C       172.16.10.0/28 is directly connected, Vlan10
L       172.16.10.2/32 is directly connected, Vlan10
C       172.16.20.0/28 is directly connected, Vlan20
L       172.16.20.2/32 is directly connected, Vlan20
C       172.16.99.0/28 is directly connected, Vlan99
L       172.16.99.2/32 is directly connected, Vlan99
S*   0.0.0.0/0 [1/0] via 10.30.10.2
```

Static default route selected because AD 1 is lower than OSPF AD 110.

WAN Architecture



WAN Routing Summary

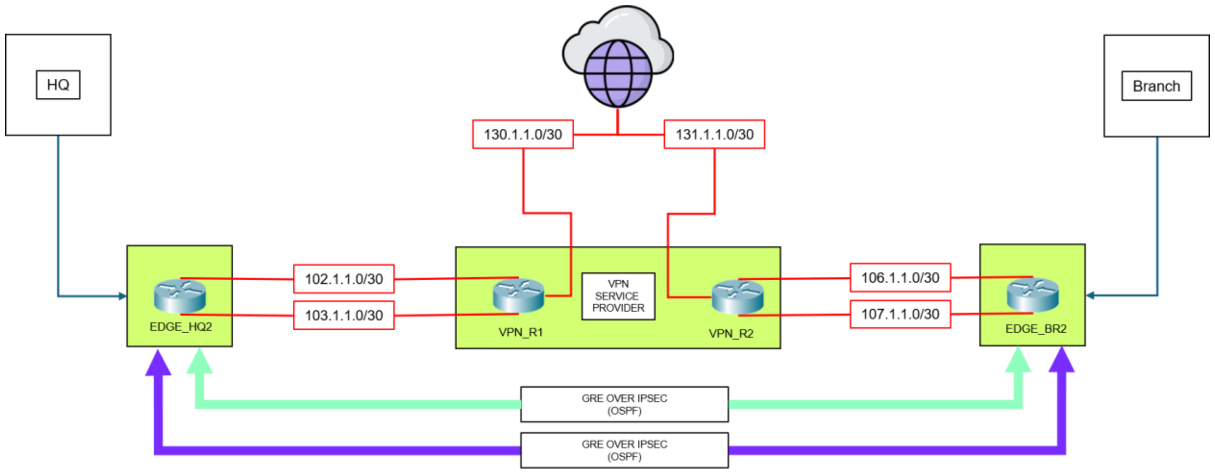
Path	Network	Connected Devices	Purpose	Routing
HQ Internet Primary	100.1.1.0/30	EDGE_HQ1 ↔ TO_INTERNET1	Internet Access (Public NAT prefixes advertised via static routes)	Static Route (Primary)
HQ Internet Backup	101.1.1.0/30			Floating Static Route
Branch Internet Primary	104.1.1.0/30	EDGE_BR1 ↔ TO_INTERNET2		Static Route (Primary)
Branch Internet Backup	105.1.1.0/30			Floating Static Route
HQ VPN	102.1.1.0/30	EDGE_HQ2 ↔ VPN_R1	GRE/IPsec Transport	Static Route (Inter-site routing is handled by GRE Tunnels via OSPF)
	103.1.1.0/30			
Branch VPN	106.1.1.0/30	EDGE_BR2 ↔ VPN_R2		
	107.1.1.0/30			
ISP Transit	120.1.1.0/30	ISP Routers ↔ Internet Routers	Internet Connectivity (Public NAT prefixes advertised via static routes)	Static Routes
	121.1.1.0/30			
VPN Transit	130.1.1.0/30	VPN Routers ↔ Internet Routers	Tunnel Transport	Static Routes
	131.1.1.0/30			
Public Service	8.8.8.0/30	INTERNET ↔ google.dns	Public Service Access	Static Routes

GRE Tunneling

GRE Tunneling was implemented to provide a logical point-to-point connection between sites and to support OSPF route exchange, enabling dynamic routing across the WAN. Security is provided separately through **IPsec** encryption.

TUNNEL OVERLAY TRANSPORT				
Tunnel	**Network**	**HQ Tunnel IP**	**Branch Tunnel IP**	**Routing Protocol**
Tunnel 1	10.0.1.0/30	10.0.1.1	10.0.1.2	OSPF
Tunnel 2	10.0.2.0/30	10.0.2.1	10.0.2.2	OSPF

TUNNEL UNDERLAY TRANSPORT			
Tunnel	**HQ WAN Network**	**Branch WAN Network**	**Transport**
Tunnel 1	102.1.1.0/30	106.1.1.0/30	GRE over IPsec
Tunnel 2	103.1.1.0/30	107.1.1.0/30	GRE over IPsec



IPsec Parameters

IPSEC PHASE 1 (IKE/ISAKMP) CONFIGURATION						
Tunnel	**Local Peer**	**Remote Peer**	**Encryption**	**Authentication**	**DH Group**	**Pre-Shared Key**
Tunnel 1	102.1.1.1 (HQ)	106.1.1.1 (Branch)	AES-128	Pre-Shared Key	Group 5	AJ123
Tunnel 2	103.1.1.1 (HQ)	107.1.1.1 (Branch)				AJ123

IPSEC PHASE 2 (IPSEC SA) CONFIGURATION					
Tunnel	**Interesting Traffic (ACL)**	**Transform Set**	**Encryption**	**Integrity**	**Crypto Map**
Tunnel 1	(GRE) 102.1.1.1 ↔ 106.1.1.1	T1-SET	ESP-AES	ESP-SHA-HMAC	T1-MAP
Tunnel 2	(GRE) 103.1.1.1 ↔ 107.1.1.1	T2-SET			T2-MAP

CRYPTO MAP APPLICATION		
Tunnel	**HQ Interface**	**Branch Interface**
Tunnel 1	G0/1/0	G0/1/0
Tunnel 2	G0/2/0	G0/2/0

SERVICES – DHCP & DNS (HQ)

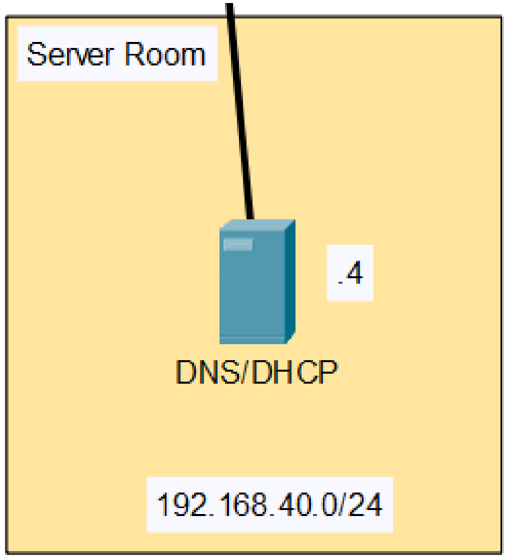
A combined **DHCP** and **DNS** server is deployed in HQ to centralize services. This setup simplifies administration and ensures consistent and convenient IP management across all offices.

DHCP Pool Table

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User
Office_A	192.168.10.1	192.168.40.4	192.168.10.11	255.255.255.0	245
Office_B	192.168.20.1	192.168.40.4	192.168.20.11	255.255.255.0	245
Office_C	192.168.30.1	192.168.40.4	192.168.30.11	255.255.255.0	245
AP_A	192.168.110.1	192.168.40.4	192.168.110.11	255.255.255.0	245
AP_B	192.168.120.1	192.168.40.4	192.168.120.11	255.255.255.0	245
AP_C	192.168.130.1	192.168.40.4	192.168.130.11	255.255.255.0	245

DNS Records Table

No.	Name	Type	Detail
0	ajtadeo.com	A Record	203.0.113.201
1	dns.google	A Record	8.8.8.8
2	portal.hq	A Record	172.16.10.8



VLAN 40

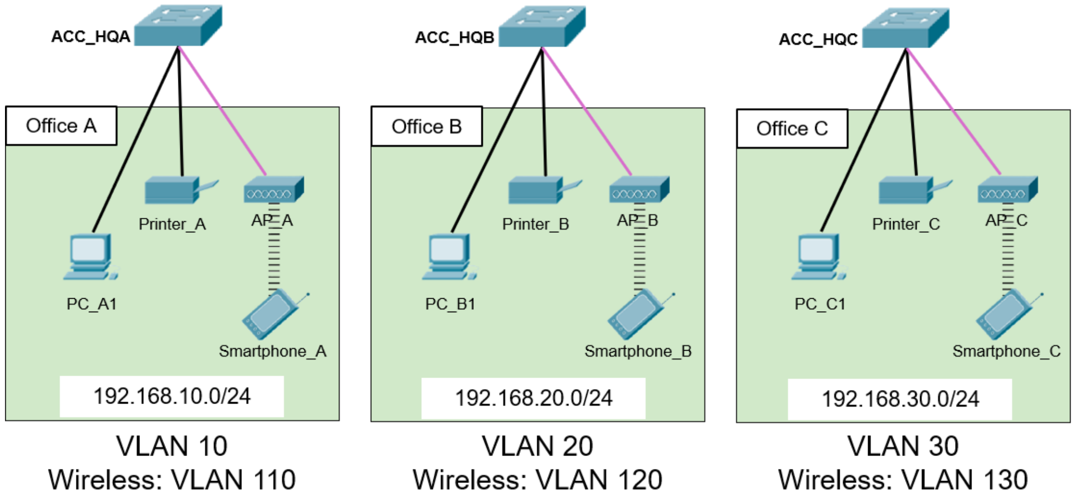
The **SVIs** on DIST_HQ1 and DIST_HQ2 are configured with `ip helper-address 192.168.40.4` command, allowing them to function as **DHCP Relay Agents**.

These agents listen for broadcast requests on end-user VLANs and send them to the central server as unicast packets.

SERVICES – Wireless Connectivity (HQ)

The offices use AP_PT-N access points operating on the IEEE 802.11n standard, supporting both **2.4 GHz** and **5 GHz** bands. Although IEEE 802.11ac provides higher performance, 802.11n was selected to ensure compatibility with devices that support only the 2.4 GHz band.

Access Point	VLAN	SSID	Authentication	Passphrase
AP_A	110	Office_A	WPA2-PSK	password123
AP_B	120	Office_B	WPA2-PSK	password123
AP_C	130	Office_C	WPA2-PSK	password123



SERVICES – FTP & HTTPS (Branch)

Portal Server (HTTPS)

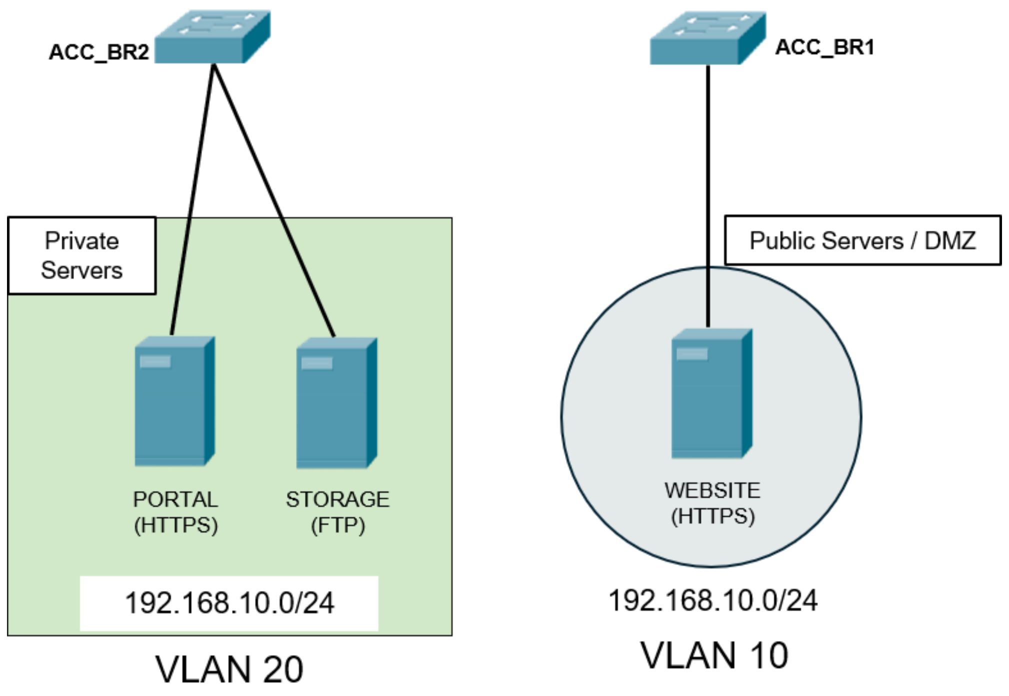
- Provides internal web services accessible only through the GRE over IPsec tunnels.
- The portal is intended for enterprise users and is not exposed to the Internet.

Storage Server (FTP)

- Provides centralized file storage and transfer services for enterprise users.
- FTP access is restricted to internal networks and protected by firewall policies.

Website Server (HTTPS)

- Hosts the organization's public-facing website.
- External users access the service through the translated public address (**203.0.113.201**) while the private address remains hidden.
- Direct access to the private IP is restricted to the Network Administrator.



SERVICES – FTP & HTTPS (Branch)

Portal Server (HTTPS)

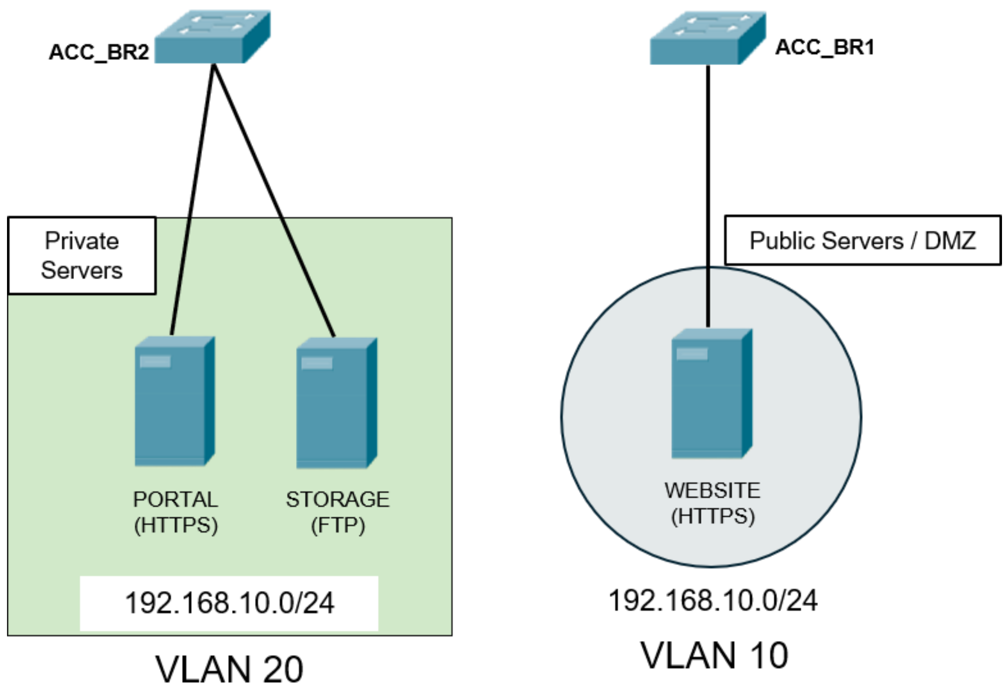
- Provides internal web services accessible only through the GRE over IPsec tunnels.
- The portal is intended for enterprise users and is not exposed to the Internet.

Storage Server (FTP)

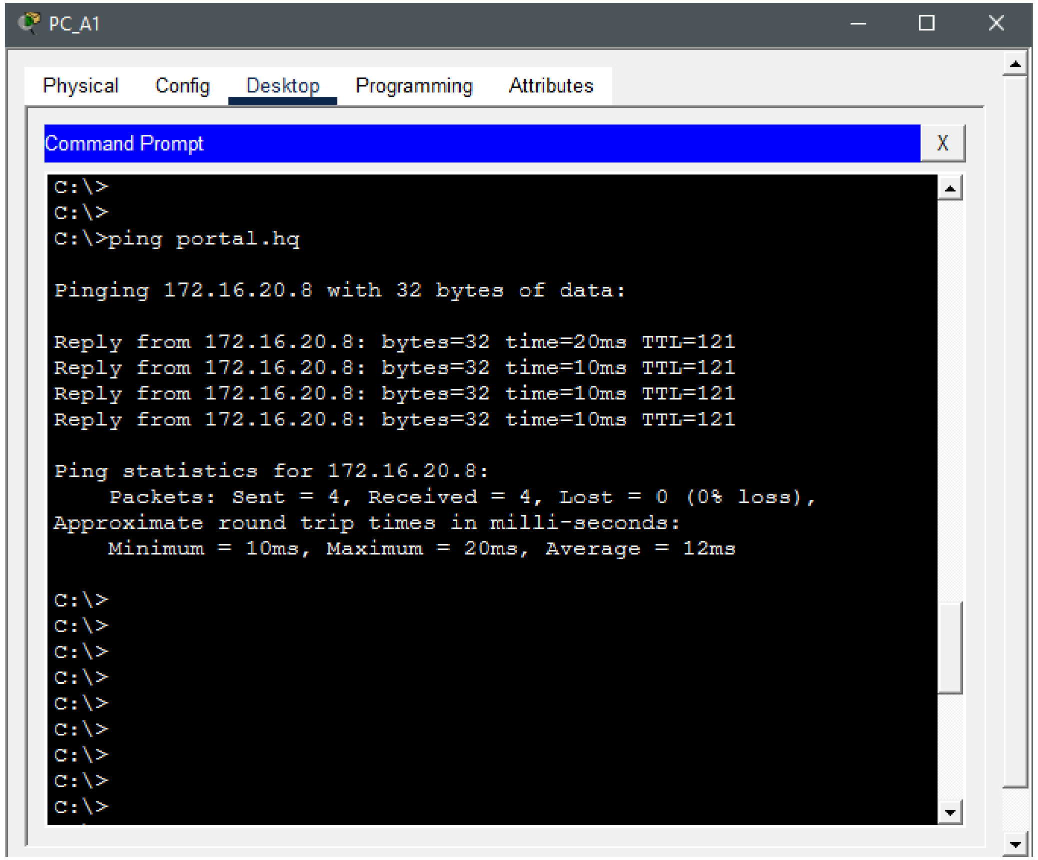
- Provides centralized file storage and transfer services for enterprise users.
- FTP access is restricted to internal networks and protected by firewall policies.

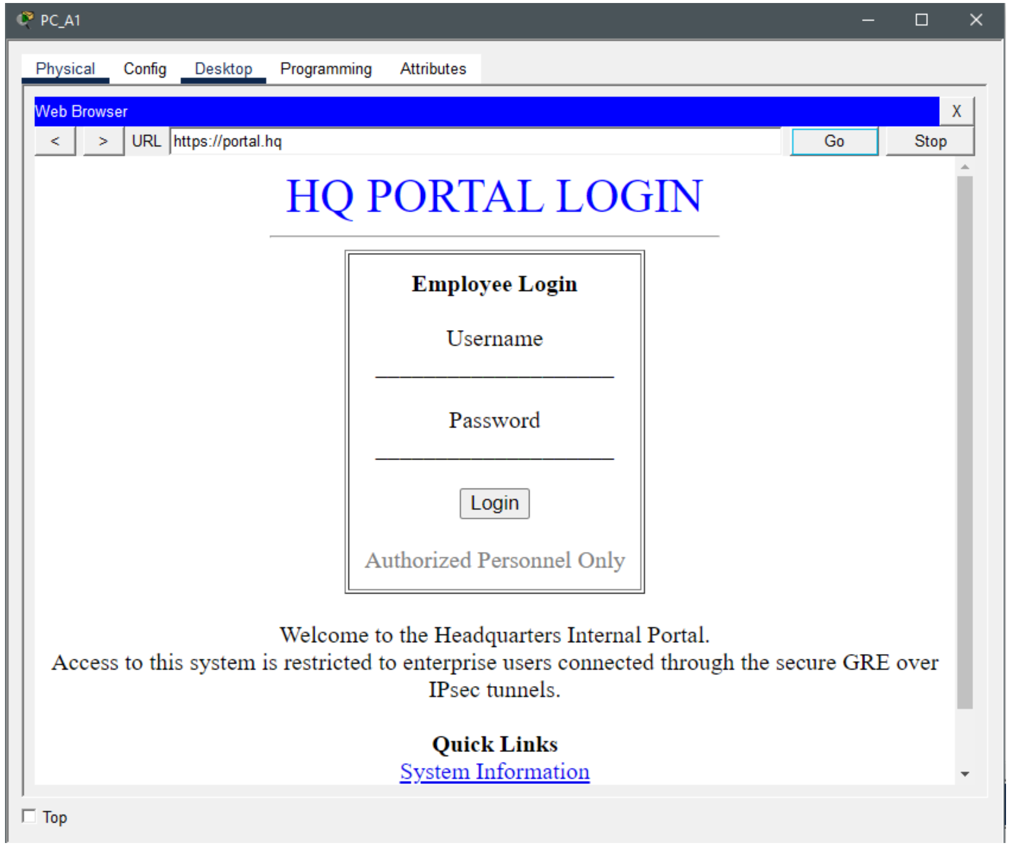
Website Server (HTTPS)

- Hosts the organization's public-facing website.
- External users access the service through the translated public address (**203.0.113.201**) while the private address remains hidden.
- Direct access to the private IP is restricted to the Network Administrator.



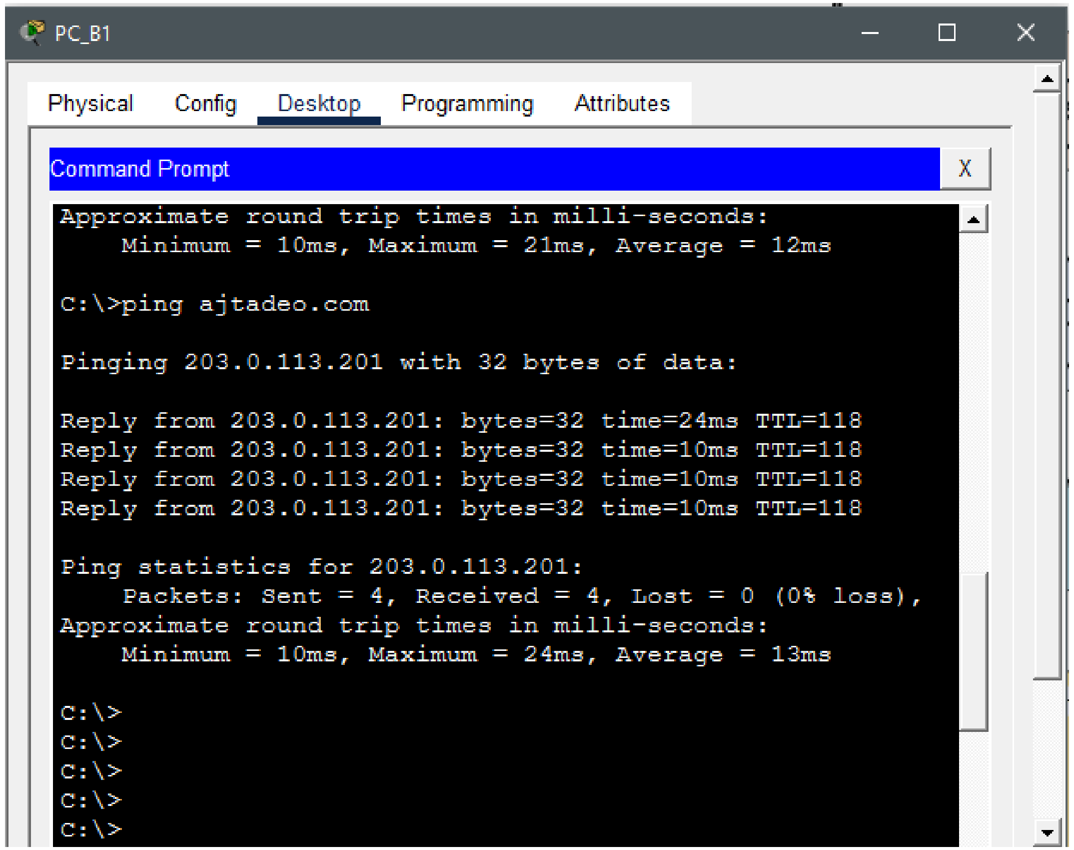
Verification – PORTAL(HTTPS)

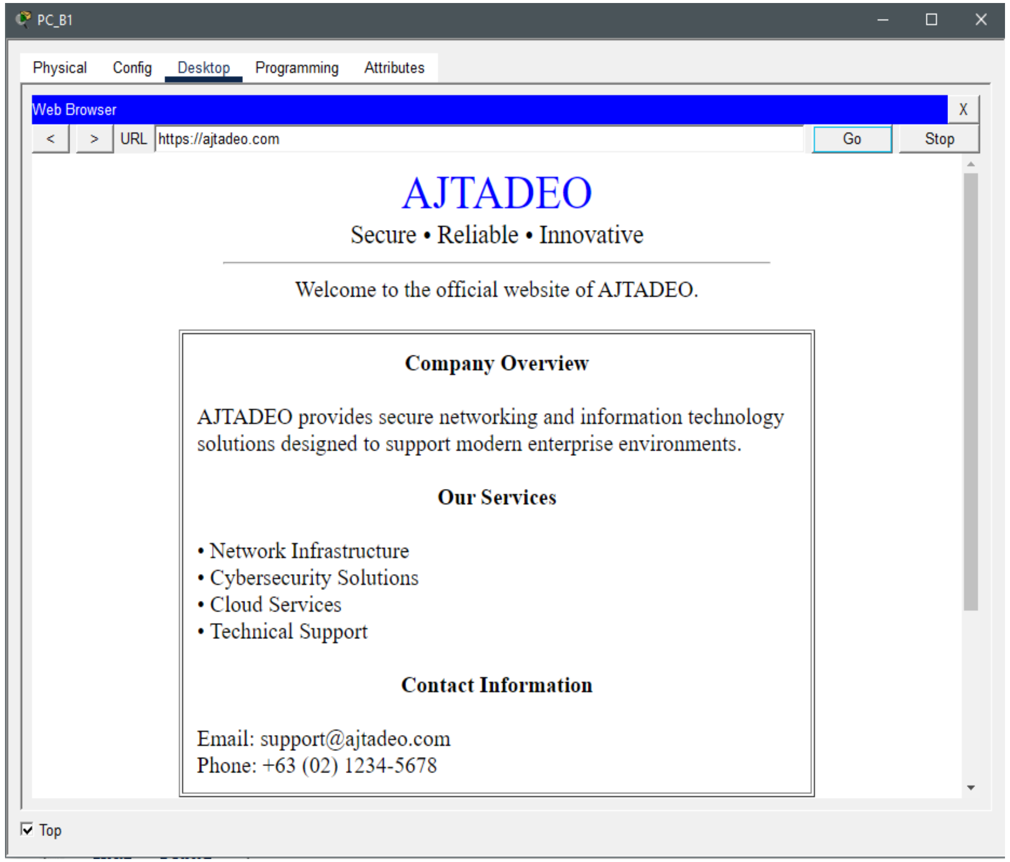




Note: Occasional timeouts on initial ping attempts are expected due to ARP resolution and do not indicate connectivity issues.

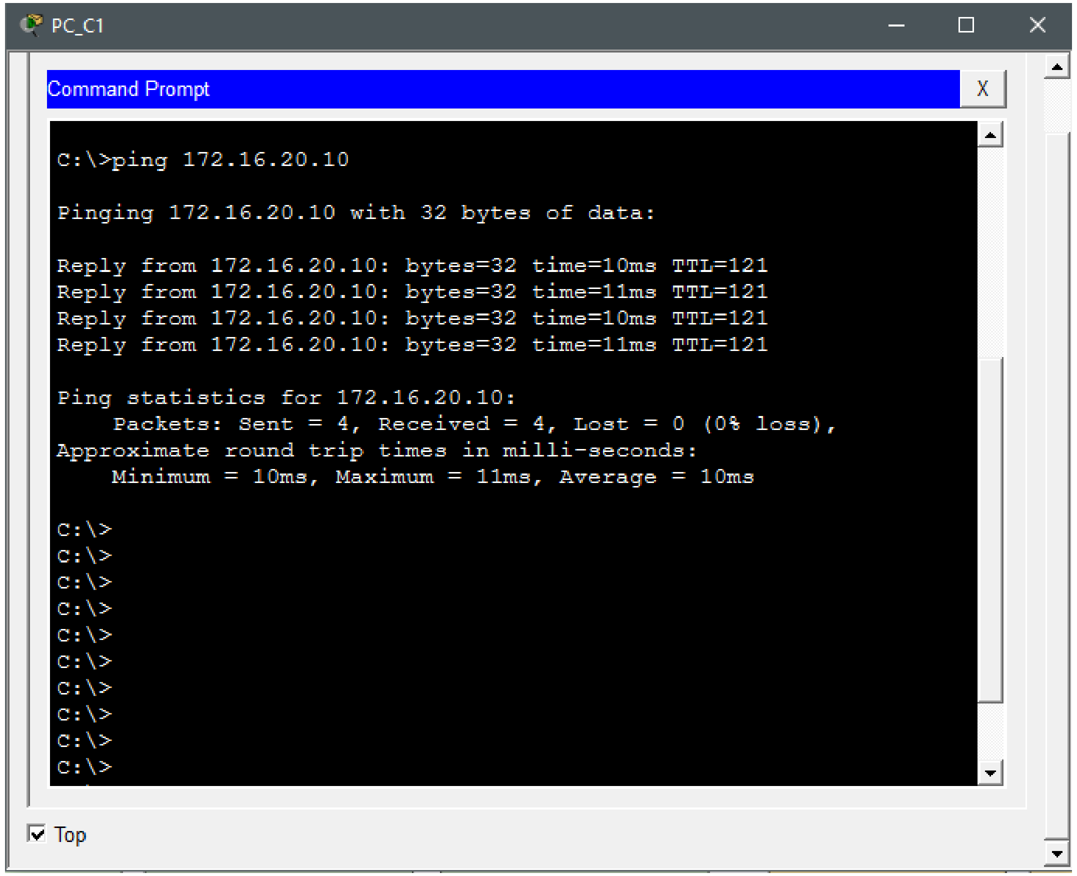
Verification – WEBSITE (HTTPS)

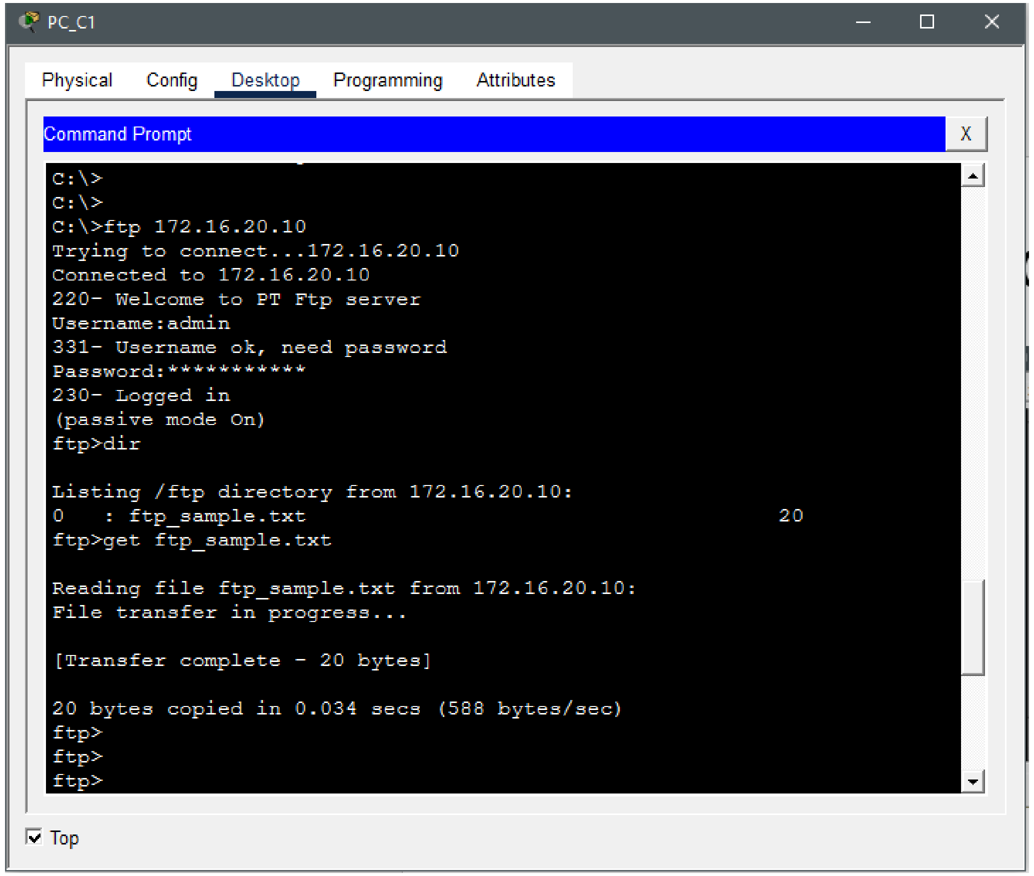




Note: Occasional timeouts on initial ping attempts are expected due to ARP resolution and do not indicate connectivity issues.

Verification – STORAGE (FTP)





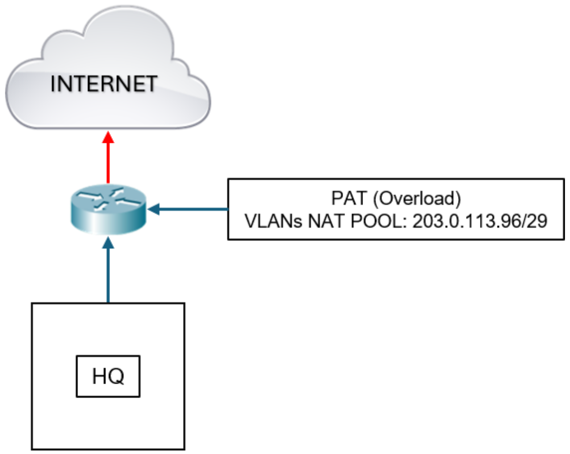
Note: Occasional timeouts on initial ping attempts are expected due to ARP resolution and do not indicate connectivity issues.

Network Address Translation

CONFIGURED ON EDGE_HQ1					
NAT Type	**ACL**	**Private IP (Inside Local)**	**Public IP (Inside Global)**	**Inside Interface**	**Outside Interface**
PAT (Overload) via NAT Pool	10	All VLANs except Management	203.0.113.96/29 (203.0.113.97 - 203.0.113.102)	Port-Channel 1 (G0/0, G0/1)	G0/1/0
		All VLANs except Management			G0/2/0

NAT Verification

```
EDGE_HQ1#  show ip nat translations
Pro  Inside global      Inside local        Outside local       Outside global
icmp 203.0.113.97:1     192.168.30.11:1     8.8.8.8:1           8.8.8.8:1
icmp 203.0.113.97:2     192.168.30.11:2     8.8.8.8:2           8.8.8.8:2
icmp 203.0.113.97:3     192.168.30.11:3     8.8.8.8:3           8.8.8.8:3
icmp 203.0.113.97:4     192.168.30.11:4     8.8.8.8:4           8.8.8.8:4
icmp 203.0.113.97:5     192.168.30.11:5     8.8.8.8:5           8.8.8.8:5
icmp 203.0.113.97:6     192.168.30.11:6     8.8.8.8:6           8.8.8.8:6
icmp 203.0.113.97:7     192.168.30.11:7     8.8.8.8:7           8.8.8.8:7
icmp 203.0.113.97:8     192.168.30.11:8     8.8.8.8:8           8.8.8.8:8
```

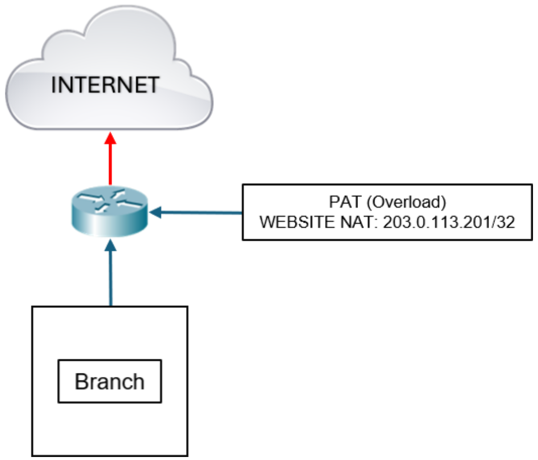


Network Address Translation

<table>
<tr><th colspan="6">CONFIGURED ON EDGE_BR1</th></tr>
<tr><th>NAT Type</th><th>Service</th><th>Private IP (Inside Local)</th><th>Public IP (Inside Global)</th><th>Inside Interface</th><th>Outside Interface</th></tr>
<tr><td rowspan="2">PAT (Overload)
via Static NAT</td><td rowspan="2">WEBSITE</td><td rowspan="2">172.16.10.8</td><td rowspan="2">203.0.113.201 /32
(Publicly Accessible)</td><td rowspan="2">Port-Channel 1
(G0/0, G0/1</td><td>G0/1/0</td></tr>
<tr><td>G0/2/0</td></tr>
<tr><td rowspan="2">NON-NAT
(STRICTLY Internal Only)</td><td>PORTAL</td><td>172.16.20.8</td><td rowspan="2">X</td><td rowspan="2">X</td><td rowspan="2">X</td></tr>
<tr><td>FTP</td><td>172.16.20.10</td></tr>
</table>

NAT Verification

```
EDGE_BR1# show ip nat translations
Pro  Inside global      Inside local       Outside local      Outside global
icmp 203.0.113.201:2    172.16.10.8:2      8.8.8.8:2          8.8.8.8:2
icmp 203.0.113.201:3    172.16.10.8:3      8.8.8.8:3          8.8.8.8:3
icmp 203.0.113.201:4    172.16.10.8:4      8.8.8.8:4          8.8.8.8:4
icmp 203.0.113.201:5    172.16.10.8:5      8.8.8.8:5          8.8.8.8:5
icmp 203.0.113.201:6    172.16.10.8:6      8.8.8.8:6          8.8.8.8:6
icmp 203.0.113.201:7    172.16.10.8:7      8.8.8.8:7          8.8.8.8:7
icmp 203.0.113.201:8    172.16.10.8:8      8.8.8.8:8          8.8.8.8:8
---  203.0.113.201      172.16.10.8        ---                ---
```



SECURITY - ASA Firewall Configuration

Firewall	Inside Interface(s)	Outside Interface	Security Levels
ASA-HQ1	10.20.10.2/30	10.20.10.5/30	100 (IN) / 0 (OUT)
ASA-HQ2	10.20.20.2/30	10.20.20.5/30	100 (IN) / 50 (OUT)
ASA-BR1	10.30.10.2/30 10.30.10.10/30	10.30.10.5/30	100 (IN) / 0 (OUT)
ASA-BR2	10.30.20.2/30 10.30.20.10/30	10.30.20.5/30	100 (IN) / 50 (OUT)

SECURITY LEVEL LEGEND	
Security Level	**Purpose**
100	Fully Trusted (Internal Network)
50	Semi-Trusted (Site-to-Site Communication)
0	Untrusted (Internet Connectivity)

SECURITY – ACLs Enforcement Architecture

Device	Security Function
EDGE_HQ1	Internet edge filtering and return-traffic validation
ASA-HQ1	Headquarters Internet perimeter protection
EDGE_HQ2	GRE/IPsec tunnel protocol protection
ASA-HQ2	Controls branch-to-HQ access through tunnels
Core / Distribution ACLs	Enforce Management VLAN isolation
EDGE_BR1	Public website traffic filtering
ASA-BR1	DMZ protection for the Website server
EDGE_BR2	Branch VPN tunnel protection
ASA-BR2	Protects Portal and Storage servers

ACL Security Policy Overview

Internal Segmentation

- User VLANs are isolated from Management VLANs.
- Management VLAN can access all enterprise networks.
- Headquarters users can access internal branch services through GRE/IPsec tunnels.

Branch Service Access

- Portal (HTTPS) and Storage (FTP) are accessible from HQ through GRE/IPsec tunnels. Management VLAN may access all branch networks and services for administration and troubleshooting.
- HQ networks may initiate communication toward branch resources.
- Branch networks and servers cannot initiate communication toward HQ networks.

Public Website Access

- WEBSITE is published through NAT (203.0.113.201).
- Internet users access the website using HTTPS.
- HQ’s VLAN 99 (Management) may access the website using its private IP address.

General Security Principle

- Explicitly permitted traffic is allowed. All other traffic is denied.

Access Control Summary

HQ User VLANs (10,20,30,40,110,120,130)
- ✓ Internet access (through NAT)
- ✗ Access to VLAN 98 and VLAN 99
- ✓ HTTPS to Portal (172.16.20.8)
- ✓ FTP to Storage (172.16.20.10)
- ✓ Ping Portal
- ✓ Ping Storage
- ✗ Private access to Website (172.16.10.8)

Management VLAN (192.168.99.0/28)
- ✗ Internet access (no NAT)
- ✓ Access to all HQ VLANs
- ✓ HTTPS to Portal
- ✓ FTP to Storage
- ✓ SSH to all Network Devices
- ✓ Ping all enterprise networks
- ✓ HTTPS to Website private IP (172.16.10.8)

Internet Users
- ✓ HTTPS to 203.0.113.201
- ✓ Ping 203.0.113.201
- ✗ Portal access
- ✗ FTP access
- ✗ Management access

Branch VLAN 10 (Website DMZ)
- ✗ Initiate ping toward HQ
- ✗ Access Branch VLAN 20
- ✗ Access Branch VLAN 99
- ✓ Reply traffic only

Branch VLAN 20 (Portal/Storage)
- ✗ Initiate ping toward HQ
- ✗ Access Branch VLAN 10
- ✗ Access Branch VLAN 99
- ✓ Reply traffic only

Layer 2 Security – Port Security

Enabled On

- HQ Access Switches
- Branch Access Switches

Configuration

- Sticky MAC learning enabled on end-user ports
- Maximum of 1 MAC address permitted on user-facing interfaces
- Restrict violation mode configured
- Access point interfaces configured with a maximum of 30 MAC addresses
- Sticky MAC learning disabled on access point ports to accommodate multiple wireless clients

Purpose

- Prevent unauthorized device connections
- Mitigate MAC flooding attacks
- Limit end-host spoofing
- Support multiple wireless clients connected through access points

Layer 2 Security – DHCP Snooping

HEADQUARTERS			
Enabled On	**Protected VLANs**	**Trusted Interfaces**	**Purpose**
ACC_HQA	10, 99, 110	Uplink Ports (G0/1 – G0/2)	• Prevent rogue DHCP Servers • Prevent DHCP starvations attacks • Build DHCP binding database
ACC_HQB	20, 99, 120		
ACC_HQC	30, 99, 130		
ACC_HQD	40, 99	Uplink Ports (G0/1 – G0/2) Server Interface (F0/1)	
ACC_HQE	99	Uplink Ports (G0/1 – G0/2)	
BRANCH			
Enabled On	**Protected VLANs**	**Trusted Interfaces**	**Purpose**
ACC_BR1	10, 99	Uplink Ports (G0/1 – G0/2)	Prevent rogue DHCP Servers
ACC_BR2	20, 99		

Layer 2 Security – Dynamic ARP Inspection

Enabled On
- Same VLANs protected by DHCP Snooping

Trusted Interfaces
- Uplink Ports
- DHCP Server Port

Validation
- Source MAC
- Destination MAC
- IP-to-MAC Binding

Purpose
- Prevent ARP spoofing and ARP poisoning attacks
- Validate IP-to-MAC bindings
- Protect host-to-host communication

Layer 2 Security – Disabling Unused Interfaces

Unused interfaces were administratively disabled and isolated to prevent

- Unauthorized device connections
- Rogue switch installations
- MAC flooding attacks
- VLAN hopping attempts
- Accidental network access

This control was implemented on the following network devices:

- Edge Routers
- ASA Firewalls
- Core Switches
- Distribution Switches
- Access Switches

ALL UNUSED PORTS WERE ASSIGNED TO VLAN 999 (UNUSED_PORTS)

Management VLAN Configuration

HEADQUARTERS			
Device	**Management IP**	**Default Gateway**	**Note**
ACC_HQA	192.168.98.4/28	192.168.98.1	SVIs were configured for remote management
ACC_HQB	192.168.98.5/28		
ACC_HQC	192.168.98.6/28		
ACC_HQD	192.168.99.4/28	192.168.99.1	
ACC_HQE	192.168.99.5/28		
BRANCH			
Device	**Management IP**	**Default Gateway**	**Note**
ACC_BR1	172.16.99.4/28	172.16.99.1	SVIs were configured for remote management
ACC_BR2	172.16.99.5/28		

SSH Configuration for Network Devices

Devices Covered

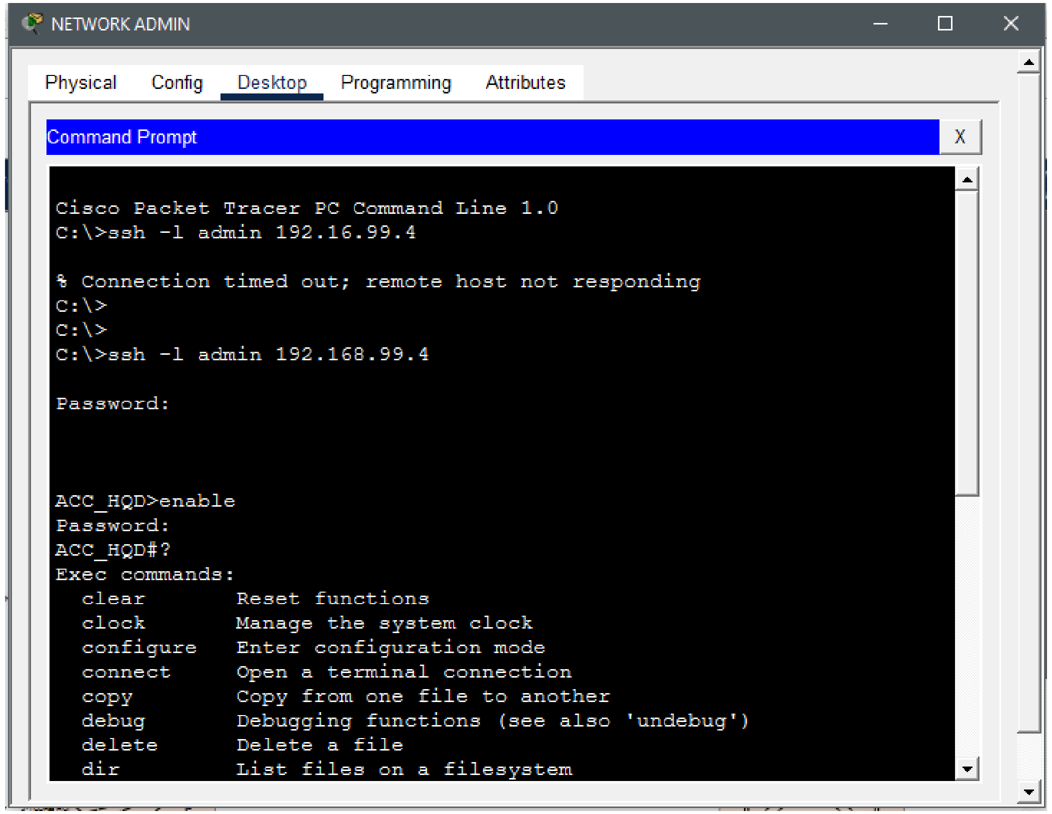
- Edge Routers
- Core Switches
- ASA Firewalls
- Distribution Switches
- Access Switches

Security Features

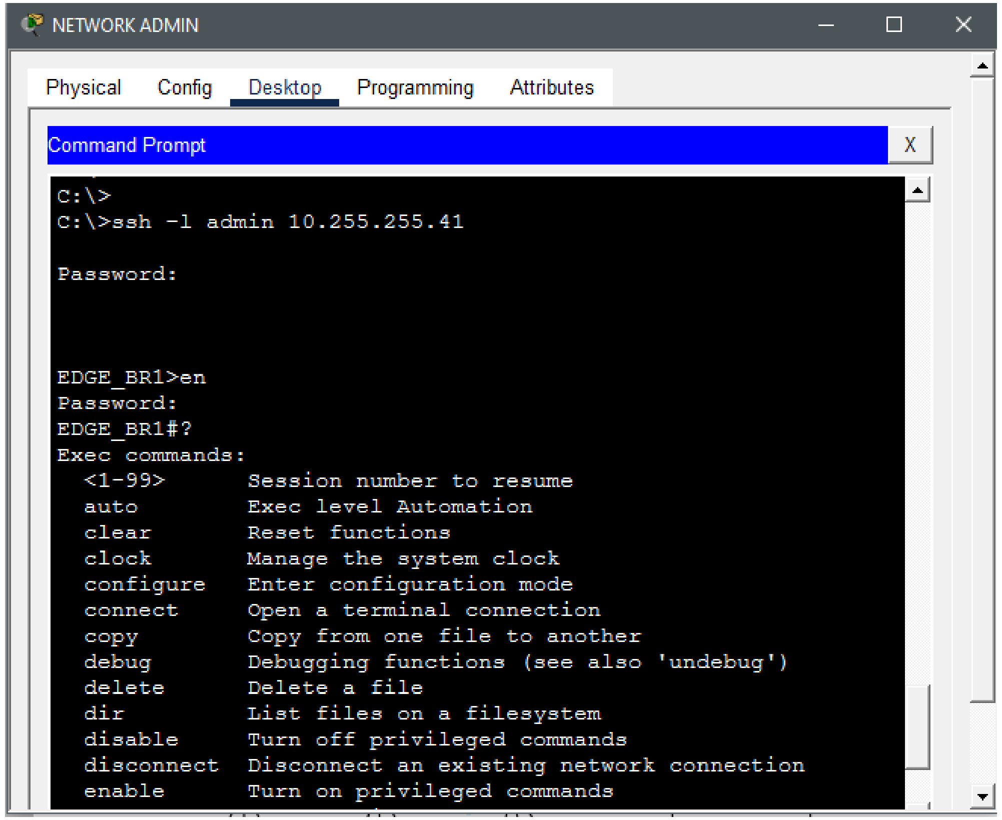
- Local user authentication
- SSH version 2 enabled
- RSA encryption keys generated
- Access restricted through SSH-MGMT ACL

Parameter	HQ	Branch
Username	admin	admin
Enable Secret	password123	password123
Domain Name	hq.local	branch.local
SSH Version	2	2
RSA Key Length	2048-bit	2048-bit
Console Timeout	30 min	30 min
Authentication	Local	Local
Access Method	Local / SSH	Local / SSH
Authorized SSH Host	192.168.99.10	192.168.99.10

Verification - SSH



SSH ACC_HQD (HQ)



SSH EDGE_BR1 (BRANCH)

Conclusion

Implemented Features

- Multi-site enterprise architecture
- GRE over IPsec VPN connectivity
- OSPF and BGP dynamic routing
- NAT and PAT services
- Public Website, Portal, FTP, DHCP, DNS and WLAN services
- ASA firewall protection
- ACL-based traffic control
- Layer 2 security (Port Security, DHCP Snooping, DAI)
- Management VLAN isolation
- Secure SSH administration
- Verification of connectivity and services